

Zonglin Di

CONTACT INFORMATION	Tongji University, 1239, Siping Road Yangpu District, Shanghai, P. R. China, 201800	(+86) 185-1669-7168 1452640@tongji.edu.cn
RESEARCH INTEREST	Geographic knowledge discovery with deep learning, Automatic control with graph neural network (GNN), Weakly-supervised learning	
EDUCATION	Tongji University , Shanghai, China B.E., Control Science and Engineering, Sept 2014 to June 2019 • GPA: 4.25/5.0 • Related Courses: Higher Mathematics, Linear Algebra, Possibility Theory & Statistics, Pattern Recognition, Object-Oriented Programming, Database Systems, Operating System, Compiler Principle, Digital Signal Processing, Digital Image Processing, Analog/Digital Circuits, Signal & Systems	
RESEARCH EXPERIENCE	Research Assistant of Deep Learning Laboratory June 2018 to Present • Perform large-scale roadmap extraction training on 40+ GPUs distributed training and rough labeling to study the limitation of weakly-supervised learning in semantic segmentation. • Improve the performance of road extraction using deep learning and GPS data, allowing 5% increase in accuracy and 40% boost in generalization ability for new area prediction. • Advisor: Prof. Yin Wang, Ph.D. of University of Michigan. Research Intern in Intel Asia & Pacific R & D Oct 2017 to Mar 2018 • Improved the performance of existing semantic segmentation algorithms for removing human portrait from an image and patching the vacant area with variational auto encoders (VAE). • Directed by Mr. Jiajie Yao Research Assistant of State Lab. of Embedded System & Service Computing, Ministry of Education Dec 2014 to Sept 2019 • Developed a novel method to facilitate the detection of Internet addiction disorder (IAD) so as to reduce the subjectivity compared to the traditional methods; completed a systematic test, showing the new model's capability to identify optimal feature combination in four questionnaires, with the accuracy of IAD labeling reaching 96.2%. • Advisor: Prof. Asoke K. Nandi, <i>IEEE Fellow</i> Research Assistant of State Key Lab. of Management & Control for Complex Systems, Ministry of Education, Sept 2016 to Sept 2019 • Develop a hybrid method for binary and multi-class imbalance classification based on blending support vector machine (SVM) and clustering algorithms and using density peak as the metric; incorporate data distribution and data imbalance simultaneously for improving the imbalance classification of SVM. • Supervisors: Prof. Mengchu Zhou, <i>IEEE Fellow</i> & Prof. Qi Kang, <i>IEEE Senior Fellow</i>	
PUBLICATIONS	1. Zonglin Di, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, "Internet Addiction Disorder Detection of Chinese College Students Using Several Personality Questionnaire Data and Support Vector Machine" <i>Addictive Behavior Report, Elsevier, 2019</i> 2. Zonglin Di, Qi Kang, Daogang Peng and Mengchu Zhou, "Density Peak based Pre-clustering Support Vector Machine for Multi-class Imbalanced Classification" <i>IEEE Conference on System Man and Cybernetics (SMC) 2019</i>	

3. Tao Sun, **Zonglin Di**, Pengyu Che, Chun Liu and Yin Wang , “Leveraging Crowdsourced GPS Data for Road Extraction from Aerial Imagery”
IEEE Conference on Computer Vision and Pattern Recognition (CVPR) 2019
4. Tao Sun, **Zonglin Di** and Yin Wang, “Combining Satellite Imagery and GPS Data for Road Extraction”
2nd ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI), 2018
5. **Zonglin Di**, Siya Yao, Qi Kang and Mengchu Zhou, “AP-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification”
IEEE International Conference on Systems, Man, and Cybernetics (SMC), 2018
6. **Zonglin Di** and Qi Kang, “ADPclust-SVM: An Adaptive Clustering-based Support Vector Machine Algorithm for Imbalance Classification” (Abstract Accepted)
IEEE International Conference on Networking, Sensing and Control (ICNSC), 2018
7. **Zonglin Di**, Xiaoliang Gong, Jingyu Shi, Hosameldin O. A. Ahmed and Asoke K. Nandi, “Detection of Internet Addiction Disorder Based on Personality Questionnaires of Chinese College Students and SVMs”
10th International Congress on Image and Signal Processing, BioMedical Engineering and Informatics (ICSP-BMEI), 2017
8. Xiaoliang Gong, Bozhong Long, Kun Fang, **Zonglin Di**, Yichu Hou and Lei Cao, “A prediction based on clustering and personality questionnaire data for IGD risk: A preliminary work”
12th International Conference on Neural Computation, Fuzzy System and Knowledge Discovery (ICNC-FSKD), 2016

TEACHING ASSISTANT	Python Programming (10064701)	Sept 2018 to Nov 2018
	Deep Learning (10068501)	March 2019 to June 2019
	Department of Computer Science and Technology, Tongji University Supervised by: Prof. Yin Wang	
	Open Source Hardware and Programming (55010501)	Sept 2018 to Jan 2019
	School of Design and Innovation, Tongji University Supervised by: Prof. Xiaohua Sun	
INTERN EXPERIENCES	Full-Stack Software Intern in National Instruments, Shanghai	April 2018 to April 2019
	Established framework of Auto Build System in the Academic Group using React.js, Flask, and JQuery. Implemented frontend and backend within one-month timeframe independently. Directed by Mr. Dasheng Xu	
	Software Intern in National Instruments, Shanghai	July 2017 to Oct 2017
	Diadem data plug-in from German and Japanese customers using C++. Directed by Ms. Mavis Jin	
	Software Intern in Microsoft, Shanghai	July 2016 to Sept 2016
	Built ASP.NET WebSite and the back-end engine. Directed by Mr. Tarry Wang and Alex Zhang	
	Technical Marketing Intern in Jikexueyuan, Beijing	July 2015 to Sep 2015
	Solved the Android issues in the community	

SELECTED AWARDS	• The 2nd Price, National "Challenge Cup" Academic Science and Technology Competition, 2019	Nov 2019
	• Special Award, "Challenge Cup" Academic Science and Technology Competition Shanghai, 2019	June 2019
	• The 2nd Price, in Shanghai Collegiate Maker Competition	Jan 2018
	• The 3rd Prize, , China Undergraduate Mathematical Contest in Modeling (Shanghai Division)	Nov 2017
	• Social Practice Scholarship, Tongji University	Sept 2017
	• The 3rd Prize, The 5th "Taidi Cup" Data Mining Contest	Sept 2017
	• The 3rd Prize, The 15th Diligent Design Contest	July 2017
	• The 3rd Prize, The 1st "Huichuangqingchun" Culture & Creativity Project	March 2016
SELECTED PRESENTATIONS	• SMC 2019, Bari, Italy	Oct 2019
	• CVPR 2019, Long Beach, USA	June 2019
	• GeoAI 2018, Seattle, USA	Nov 2018
	• SMC 2018, Miyazaki, Japan	Oct 2018
	• ICNSC 2018, Zhuhai, China	March 2018
	• ICSP-BMEI 2017, Shanghai, China	Sept 2017
	• ICNC-FSKD 2016, Hunan, China	Aug 2016
	• Excellent Undergraduate, Tongji University	Sep 2015, 2016, 2017
SELECTED SOCIAL EXPERIENCE	IEEE CVPR 2019 Student Volunteer	June 15 to June 21, 2019
	Virtual Instruments Club	Sept 2015 to Sept 2018
	Member, Minister of Technical Department, Vice President, President	
	Collaborated with National Instruments (Great China Area) in organizing multiple on-campus events at Tongji University	
	Delivered 20+ presentations to 200+ students	
	Google Developer Group	Dec 2016 to Sept 2018
	Invited to Google Developer Days 2016 and 2017	
	Organized the Google DevFes 2017	
	Organized the Google TensorFlow Technical Salon on the application of deep learning in computer vision and natural language processing	
	Unity Unite Shanghai and Beijing	May 2017 and 2018
	Keynote and Tech Conference Track Support	
REVIEWER EXPERIENCE	• The 2th International Conference on Mechanical, Electric and Industrial Engineering	
	• ACM SIGSPATIAL 2019 (External)	